1. **What is a computer system?**

   A computer system is a combination of hardware and software that works together to process data and produce meaningful information. It performs input, processing, storage, and output operations.

2. **Why is a computer called a digital machine?**

   A computer is called a digital machine because it processes data in binary form (0s and 1s). All operations are based on digital signals.

3. **What is binary number system?**

   The binary number system uses only two digits: 0 and 1. It is the fundamental number system used by computers to represent data.

4. **What is a bit?**

   A bit (binary digit) is the smallest unit of data in a computer. It can have only two values: 0 or 1.

5. **What is ASCII and how many characters does it represent?**

   ASCII (American Standard Code for Information Interchange) is a coding system used to represent characters in computers. It represents 128 standard characters.

6. **What is logic 0 and logic 1?**

   Logic 0 represents a LOW signal (off/false), while logic 1 represents a HIGH signal (on/true) in digital systems.

7. **What is an analog signal?**

   An analog signal is a continuous signal that varies smoothly over time. It can have infinite values within a range.

8. **What is a digital signal?**

   A digital signal is a discrete signal that has only specific values, usually 0 and 1. It is used in digital electronics and computers.

9. **Give two differences between analog and digital signals.**

Analog Signals	**Digital Signals**

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Analog signals are continuous	Digital signals are discrete
Analog signals are more affected by noise	Digital signals are more reliable.

10. Which signal is more resistant to noise and why?

Digital signals are more resistant to noise because they use fixed values (0 and 1), making errors easier to detect and correct.

11. What is a logic gate?

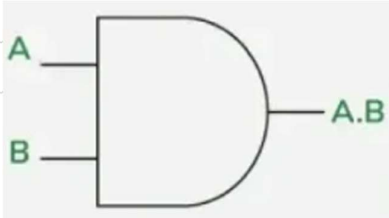
A logic gate is an electronic device that performs logical operations on one or more inputs to produce a single output.

12. Why are logic gates called building blocks of digital circuits?

Logic gates are called building blocks because all digital circuits are made by combining multiple logic gates to perform complex operations.

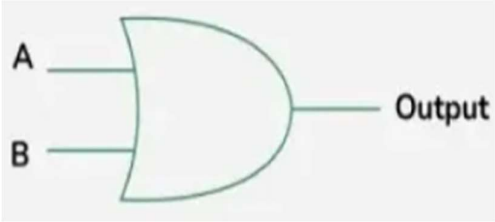
13. Define AND gate?

In and gate, when all inputs are 1 the output will be 1, else it will be 0. An AND gate have 2 or more than 2 inputs. Its logic expression for 2 input is F = A . B & for 3 it is F = A . B . C.



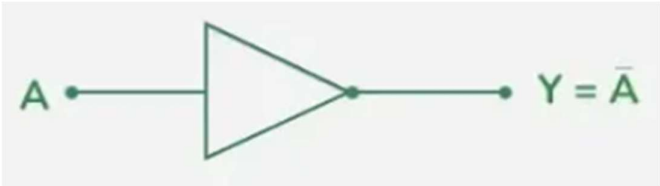
14. Define OR gate?

In or gate, if any input is 1 the output will be one, else it will be 0. It has 2 or more than 2 inputs. Its logic expression for 2 input is F = a + b & for 3 input F = a + b + c.



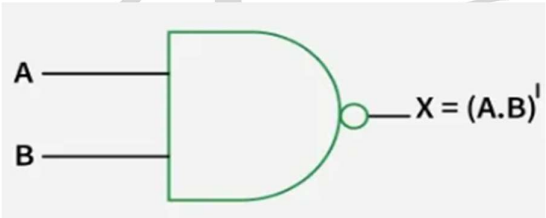
15.Define NOT gate?

NOT gate is a single input gate. If input is 1 then the output will be 0, if input is 0 the output will be 1. Its logic expression is F = X'.



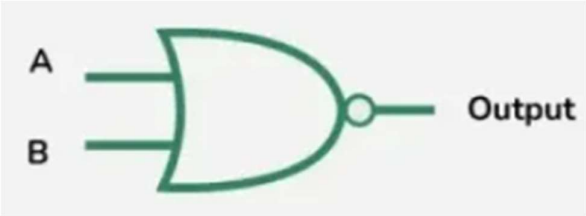
16.Define NAND gate?

NAND gate is the combination of two gates NOT and AND gate. NAND gate converts the output of and gate. Its logic expression is F = (a . b)'.



17.Define NOR gate?

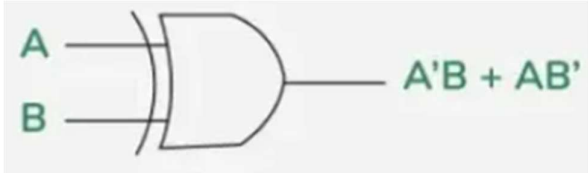
NOR gate combines two gates NOT and OR gate. NOR gate converts the output of OR gate. It logic expression is F = (A + B)'.



18.Define XOR gate?

Exclusive-OR gate has the same symbol as OR gate but with an extra curved at input. It produces output 1 only when there are different inputs,

if the inputs are same the output will be 0. Its logic expression is F = x'y + xy'.



19.Write the truth table of AND gate.

A	B	F= A.B
0	0	0
0	1	0
1	0	0
1	1	1

20.Write the truth table of OR gate.

A	B	F= A + B
0	0	0
0	1	1
1	0	1
1	1	1

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21.Write the truth table of NAND gate.

A	B	F= (A . B)'
0	0	1
0	1	1
1	0	1

1	1	0

22. Write the truth table of NOR gate.

A	B	F= (A + B)’
0	0	1
0	1	0
1	0	0
1	1	0

23. Write the truth table of NOT gate.

A	F= A’
1	0
0	1

24. Write the truth table of XOR gate.

A	B	F= A + B
0	0	0
0	1	1
1	0	1
1	1	0

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25. What is a Boolean function?

A Boolean function is a mathematical expression that uses binary variables and logical operations like AND, OR, and NOT.

26. What is Karnaugh Map (K-Map)?

A Karnaugh Map is a graphical method used to simplify Boolean expressions. It helps reduce the number of variables and logic gates.

27. Why is K-Map used for simplification?

K-Map is used to simplify Boolean expressions to reduce circuit complexity, cost, and number of logic gates used.

28. State De Morgan's Theorem.

De Morgan's Theorem states:

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

29. What are use of logic Gates?

They are used in **memory circuits** to **store binary data (0 and 1)**.

They are used in **control systems** to **control and manage signals**.

They are used in **data encoding and decoding** to **convert data for communication**.

They are used in **mathematical calculations** to **perform arithmetic operations**.

They are used in **digital signal processing** to **process and filter signals**.

They are used in **encryption and decryption** to **secure data**.

They are used in **calculator circuits** to **perform calculations quickly**.

They are used in **traffic light systems** to **control traffic flow**.

They are used in **robotics systems** to **make decisions and control movement**.

They are used in **security systems** to **control alarms and access**.

They are used in **automotive systems** to **control vehicle functions (ABS, airbags)**.

They are used in **home automation systems** to **automate devices (lights, temperature)**.

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They are used in **medical devices** to **monitor and control functions**.

They are used in **aerospace systems** to **control navigation and guidance**.

Software Development Life Cycle

30. Define Software Development Life Cycle (SDLC).

SDLC is a systematic process used to develop software in phases such as planning, analysis, design, implementation, testing, deployment, and maintenance.

31. What is meant by Feasibility Study?

A feasibility study evaluates whether a proposed software project is practical and achievable within available resources, time, and budget.

32. Define Requirement Engineering.

Requirement Engineering is the process of gathering, analyzing, documenting, and validating user requirements for a software system.

33. What is an Algorithm?

An algorithm is a step-by-step procedure or set of rules designed to solve a specific problem.

34. Define Flowchart.

A flowchart is a graphical representation of an algorithm using symbols to show the sequence of steps.

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35. What is Black Box Testing?

Black Box Testing is a testing method where the system is tested without knowing its internal code or structure.

36. What is White Box Testing?

White Box Testing is a testing method where internal code, logic, and structure of the software are tested.

37. Define Deployment Phase.

Deployment is the phase where the completed software is installed and made available for users.

38. What is Maintenance in SDLC?

Maintenance is the phase where software is updated, improved, or fixed after deployment.

39. Define Waterfall Model.

The Waterfall Model is a linear SDLC model where each phase is completed before moving to the next.

40. Define Agile Model.

The Agile Model is an iterative SDLC approach that develops software in small increments with continuous feedback.

41. What is SRS?

SRS (Software Requirement Specification) is a document that describes the functional and non-functional requirements of a system.

42. What is SDD?

SDD (Software Design Document) describes the system architecture, design details, and technical specifications.

43. Differentiate between Technical and Economic Feasibility.

Technical feasibility checks if the required technology and skills are available.

Economic feasibility checks if the project is financially viable and cost-effective.

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44. Differentiate between Verification and Validation.

Verification ensures the product is built correctly according to specifications.

Validation ensures the correct product is built to meet user needs.

45. Differentiate between Black Box Testing and White Box Testing.

Black Box Testing focuses on inputs and outputs without knowing code.

White Box Testing examines internal code and logic structure.

46. Explain the purpose of Requirement Validation.

Requirement validation ensures that documented requirements are correct, complete, and meet user needs before development starts.

47. Why is documentation important in SDLC?

Documentation provides clear guidelines for developers, helps in maintenance, and ensures smooth communication among stakeholders.

48. Differentiate between Corrective and Perfective Maintenance.

Corrective maintenance fixes errors found after deployment.

Perfective maintenance improves performance or adds new features.

49. Why is the Waterfall model not suitable for changing requirements?

Because it follows a fixed sequence, changes in later stages require revisiting earlier phases, increasing cost and time.

50. Explain the concept of sprint in Agile model.

A sprint is a short development cycle (usually 1–4 weeks) in which a small, functional part of the software is developed and delivered.

Network Topology

51. Define Network Topology.

Network topology is the physical or logical arrangement of computers and devices in a network.

52. What is a Node?

A node is any device (computer, printer, router, etc.) connected to a network.

53. Define Bus Topology.

Bus topology is a network where all devices are connected to a single main cable called the backbone.

54. Define Star Topology.

Star topology is a network where all devices are connected to a central hub or switch.

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55. What is Mesh Topology?

Mesh topology is a network where each device is connected to every other device.

56. Define Ring Topology.

Ring topology is a network where each device is connected to two others, forming a circular path.

57. What is Hybrid Topology?

Hybrid topology is a combination of two or more different network topologies.

58. What is Token Ring?

Token Ring is a network system where a special data packet called a token circulates to control data transmission.

59. Define Scalability.

Scalability is the ability of a network to expand and handle increased workload easily.

60. Define Reliability.

Reliability is the ability of a network to function properly without frequent failures.

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61. Differentiate between Physical and Logical topology.

Physical topology shows the actual physical layout of cables and devices.

Logical topology shows how data flows within the network.

62. Differentiate between Bus and Star topology.

In Bus topology, all devices share a single backbone cable.

In Star topology, all devices connect to a central hub or switch.

63. Why is Mesh topology more reliable?

Because multiple paths exist between devices, so if one link fails, data can travel through another path.

64. Why is Bus topology less reliable?

If the main backbone cable fails, the entire network stops working.

65. Explain Dual Ring topology.

Dual Ring topology uses two rings for data transmission; if one ring fails, the other continues working as a backup.

66. Why is Hybrid topology used in large networks?

It combines advantages of different topologies, providing better flexibility, scalability, and reliability for large systems.

Cloud Computing

67. Define Cloud Computing.

Cloud computing is the delivery of computing services (servers, storage, software) over the internet instead of using local computers.

68. What is IaaS?

IaaS (Infrastructure as a Service) provides virtualized computing resources like servers and storage over the internet.

69. Define PaaS.

PaaS (Platform as a Service) provides a platform and tools for developers to build and deploy applications online.

70. What is SaaS?

SaaS (Software as a Service) provides ready-to-use software applications over the internet.

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71. Define Public Cloud.

A public cloud is a cloud service available to the general public and managed by a third-party provider.

72. What is Private Cloud?

A private cloud is a cloud infrastructure used exclusively by a single organization.

73. Define Hybrid Cloud.

Hybrid cloud is a combination of public and private cloud environments working together.

74. What is Horizontal Scalability?

Horizontal scalability means adding more machines or servers to handle increased workload.

75. Define Vertical Scalability.

Vertical scalability means increasing the power (CPU, RAM) of an existing server.

76. Differentiate between IaaS and PaaS.

IaaS provides virtual hardware resources like servers and storage.

PaaS provides a development platform with tools to build applications.

77. Differentiate between Public and Private Cloud.

Public cloud is shared and available to multiple users.

Private cloud is dedicated to a single organization.

78. Why is cloud computing cost-effective?

It reduces hardware, maintenance, and infrastructure costs by using pay-as-you-go services.

79. Explain Rapid Elasticity.

Rapid elasticity means cloud resources can quickly increase or decrease based on demand.

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80. Why is redundancy important for cloud reliability?

Redundancy ensures backup systems are available, so if one system fails, another continues operating.

Cybersecurity

81. Define Cybersecurity.

Cybersecurity is the practice of protecting computer systems, networks, and data from digital attacks and unauthorized access.

82. What is Malware?

Malware is malicious software designed to damage, disrupt, or gain unauthorized access to systems.

83. Define Phishing.

Phishing is a cyberattack where attackers trick users into revealing sensitive information through fake emails or websites.

84. What is Ransomware?

Ransomware is malware that encrypts a victim's data and demands payment to restore access.

85. Define Firewall.

A firewall is a security system that monitors and controls incoming and outgoing network traffic.

86. What is Encryption?

Encryption is the process of converting readable data into coded form to prevent unauthorized access.

87. Define Cryptography.

Cryptography is the science of securing information by transforming it into an unreadable format.

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88. What is 2FA?

2FA (Two-Factor Authentication) is a security method that requires two types of verification to access an account.

89. Define Symmetric Encryption.

Symmetric encryption uses a single secret key for both encryption and decryption.

90. Define Asymmetric Encryption.

Asymmetric encryption uses two keys: a public key for encryption and a private key for decryption.

91. Differentiate between Virus and Worm.

A virus attaches itself to a file and spreads when the file is executed.

A worm spreads automatically across networks without needing a host file.

92. Differentiate between DoS and DDoS attack.

DoS (Denial of Service) attack is launched from a single system.

DDoS (Distributed Denial of Service) attack is launched from multiple systems.

93. Why is 2FA more secure than a password alone?

Because it requires an additional verification factor, making it harder for attackers to gain access even if the password is stolen.

94. Differentiate between Symmetric and Asymmetric Encryption.

Symmetric encryption uses one shared key.

Asymmetric encryption uses a pair of public and private keys.

95. Why is encryption important for data privacy?

Encryption protects sensitive data by making it unreadable to unauthorized users.

96. Explain key distribution problem in symmetric encryption.

In symmetric encryption, securely sharing the secret key between sender and receiver is difficult and risky.

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